Algorithm 1 One outer increment $T_n^* \to T_{n+1}^*$ of the coupling

Require: fields ϕ_{3D}^n, α^n, c^n, a^n; at every point the state $(\phi_1, \phi_2, \phi_0, \psi_1, \psi_2, \gamma)$ of the point model and $\sum_i \bar{\phi}_i^n = \phi_1\psi_1 + \phi_2\psi_2$

$\phi_{3D}^{n+1}, \alpha^{n+1}$: Eq. 34 with the front term driven by ∇c^n, Eq. 36 $\dot{\alpha} = k_\alpha \phi_{3D}$

c^{n+1}: $\dfrac{c^{n+1} - c^n}{d\,\Delta T} = \nabla^2 c^{n+1} - \dfrac{g}{d}\,\phi_{3D}\Big(\sum_i \bar{\phi}_i^{\,n}\Big) c^{n+1}$, $c = 1$ on the nutrient layer

a^{n+1}: the same with d_a, g_a; $a = 1$ on the outer layers once $T^* > T_a^*$

for all points with $\phi_{3D}^{n+1} \geq \phi_{\min}$ **do**

  if the point model has not started **then**

    start from the state of the colonised neighbours (or by position)

  end if

  for $k = 1, \ldots, n_{\text{sub}}$, $w = k/n_{\text{sub}}$ **do**

    $c^* = c_0^*\,[(1-w)\,c^n + w\,c^{n+1}], \quad \alpha^* = \alpha_0^*\,[(1-w)\,a^n + w\,a^{n+1}]$

    advance the point model by $s\,\Delta T/n_{\text{sub}}$ (Klempt et al. 2026, Newton) $\to \phi_i, \psi_i$

  end for

  $\sum_i \bar{\phi}_i^{n+1} = \sum_i \phi_i\psi_i$

end for

stress with $\mathbf{F}_g = \alpha^{n+1}\mathbf{I}$